

Linear models

Blended Intensive Programme
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Some References

- Fox, J. (2015). Applied Regression Analysis and Generalized Linear Models (3rd Ed). SAGE Publications, Inc.
- Fox, J. (2002). An R and S-Plus Companion to Applied Regression. SAGE.
- Gelman, A., Hill, J., & Vehtari, A. (2021). Regression and other stories. Cambridge University Press.
- Wilber, J. (2022). Linear regression. A Visual Introduction To (Almost) Everything You Should Know.
<https://mlu-explain.github.io/linear-regression/>

Linear Models

Linear Models

- A linear model is a statistical model in the form of

$$\mathbf{Y} = \mathbf{X}\beta + \epsilon$$

where $\mathbf{Y}$ is a response variable, $\mathbf{X}$ a set of predictors, ϵ the error component.

- With β we indicate the regression coefficients
- The aim of the model is to estimate these latter parameters.
- They are called linear models because the hypothesized relationship between predictors and dependent variable is assumed to be linear.
- Predictors can be quantitative or categorical.

Interpret coefficients as comparisons

- 1) Regression coefficients are commonly called *effects*, but this terminology can be misleading because they do not quantify causal effects, except for cases with specific and strong assumptions; consequently it is better to interpret it as comparisons and not as effects.
- 2) In general, regression coefficients indicate the expected differences in the values of the dependent variable for a one-unit difference in a predictor, holding the values of the other predictors constant.

For this reason it is important to control for all variables hypothesized to be related to the dependent variable.

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- **Constant variance.** The variance of the errors is the same regardless the value of X : $V(\epsilon|x_i) = \sigma^2$.
- **Normality.** The errors are normally distributed: $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$.

Simple linear regression

Simple linear regression

- A model with a single predictor X and a dependent variable Y (quantitative) is called **Simple linear regression**.

One predictor

In the case of a single predictor, the model formulation can be expressed in two ways, depending on the type of predictor:

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- 1 If X is **continuous/quantitative**

$$Y = \beta_0 + \beta_1 X + \epsilon$$

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- 1 If X is **continuous/quantitative**

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- 2 If X is **categorical**, expressed with k categories

$$Y = \beta_0 + \sum_{j=1}^{k-1} \beta_j D_j + \epsilon$$

where D are *dummy* variables that take only 0 and 1 values.

One categorical predictor

For example:

- If $k = 2$

$$Y = \beta_0 + \beta_1 D_1 + \epsilon$$

One categorical predictor

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$$Y = \beta_0 + \beta_1 D_1 + \epsilon$$

- If $k = 3$

$$Y = \beta_0 + \beta_1 D_1 + \beta_2 D_2 + \epsilon$$

One categorical predictor

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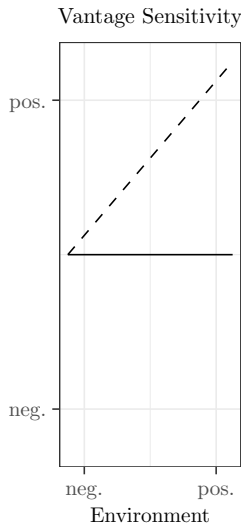
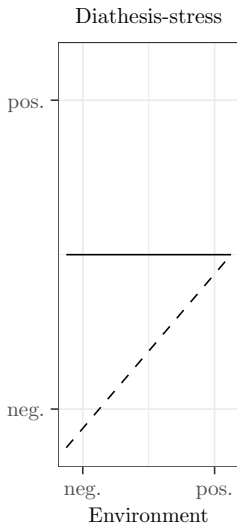
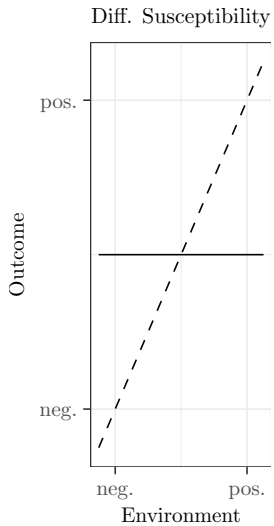
- If $k = 3$

$$Y = \beta_0 + \beta_1 D_1 + \beta_2 D_2 + \epsilon$$

- Fortunately, we don't have to worry about recoding variables because R does it automatically.
- We only need to understand the meaning of the recoding, in order to interpret the estimated parameters.

Example 1: Environmental Sensitivity

- Following recent theories, approximately 20-30% of people are **highly sensitive** to internal and external stimuli (Pluess et al., 2018; Lionetti et al., 2018; Belsky & Pluess, 2009).
- The markers of this sensitivity can be identified at the genetic, epigenetic, and neurobiological levels.
- Over the past few decades, various models have been proposed to explain specific types of individual-environment interactions.



Example 1

The `ESdata` dataset from the `ADati` R package contains data on 200 children with a mean age of 3.53 years (s.d. 0.27)¹.

The included variables are:

- `ParSty` = *Parenting Style and Dimension Questionnaire*, a measure of permissive parenting style.
- `HighSens` = level of sensitivity (1 = high).
- `ChildBeh` = *Child Behaviour Check List*, a measure of behavioural problems.
- `gender` = gender.
- `age` = age.

¹see Lionetti & al. 2019

Example 1: a first look at data

```
data( ESdata, package = "ADati")  
str( ESdata )
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'data.frame':^~I200 obs. of  5 variables:  
 $ ParSty   : num  16 9 14 14 10 11 7 12 12 6 ...  
 $ HighSens: Factor w/ 2 levels "0","1": 1 1 2 2 2 2 1 1 1 1 ...  
 $ ChildBeh: num  16 10 27 15 10 14 12 9 8 13 ...  
 $ gender   : Factor w/ 2 levels "f","m": 1 1 1 1 1 2 1 1 1 1 ...  
 $ age      : num  4.17 3.33 3.5 3.58 3.33 3.83 3.5 3.42 3.42 3.1
```

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```
head( ESdata )
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```

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head( ESdata )
```

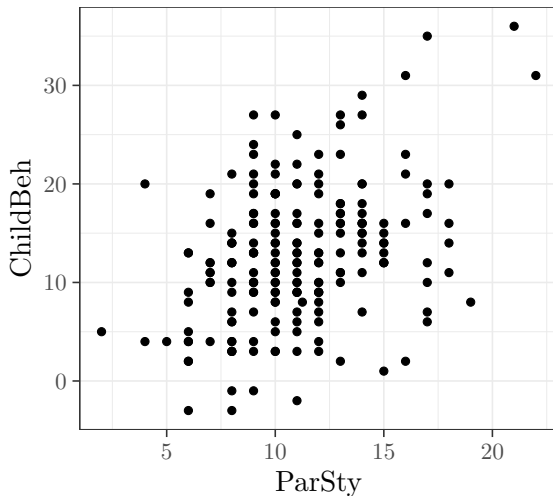
	ParSty	HighSens	ChildBeh	gender	age
1	16	0	16	f	4.17
2	9	0	10	f	3.33
3	14	1	27	f	3.50
4	14	1	15	f	3.58
5	10	1	10	f	3.33
6	11	1	14	m	3.83

One numeric predictor

We want to assess whether parenting style is a predictor of behavioural problems.

One numeric predictor

```
plot( ChildBeh ~ ParSty, data = ESdata )
```



One numeric predictor

```
m1 <- lm( ChildBeh ~ ParSty, data = ESdata )
```

One numeric predictor

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m1 <- lm( ChildBeh ~ ParSty, data = ESdata )
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```
coef( m1 )
```

(Intercept)	ParSty
3.93	0.80

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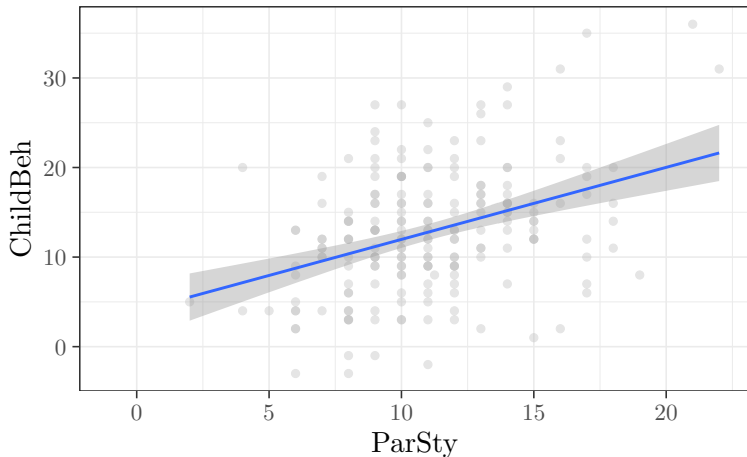
(Intercept)	ParSty
3.93	0.80

The regression equation will be

$$\hat{Y} = 3.93 + 0.8X$$

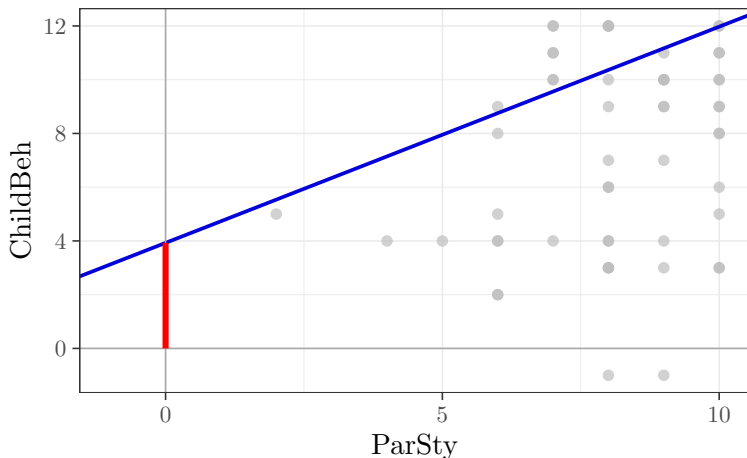
Parameters interpretation

- The points on the straight line, $\hat{Y}$, represent the expected values of **ChildBeh** in relation to the parental permissive style scores (X).



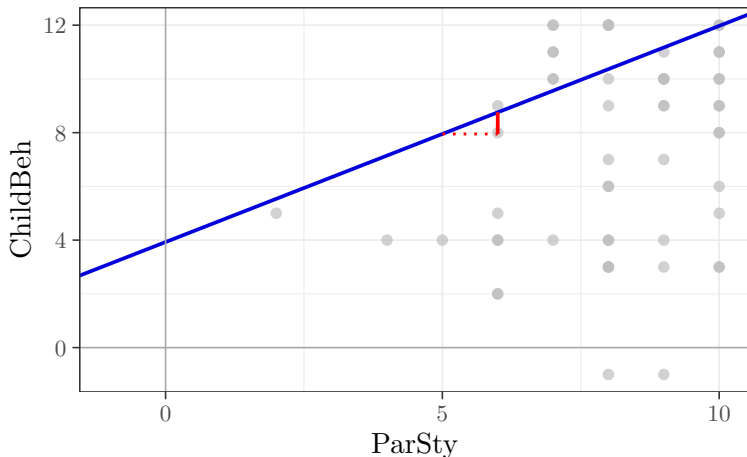
Parameters interpretation

- The intercept $b_0 = 3.93$ indicates the expected score in ChildBeh for a subject with ParSty equal to zero.



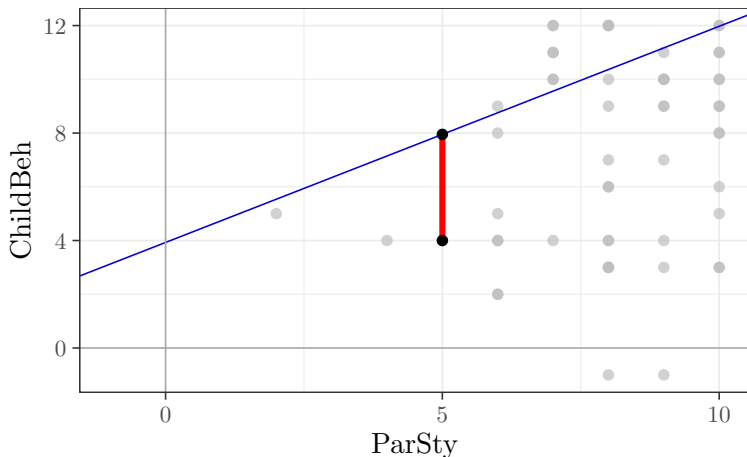
Parameters interpretation

- The regression coefficient $b_1 = 0.8$ indicates that, comparing two subjects with a difference of 1 in ParSty, we expect to observe a difference about 0.8 points in ChildBeh scores.



Residuals

- Residuals are the difference between $\hat{y}_i$, the expected value and y_i the observed one.

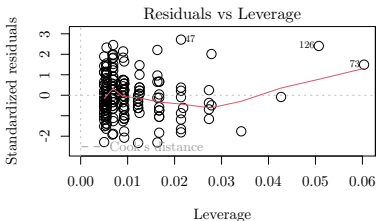
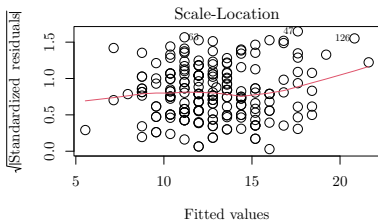
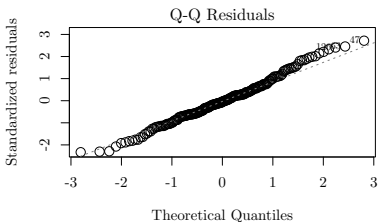
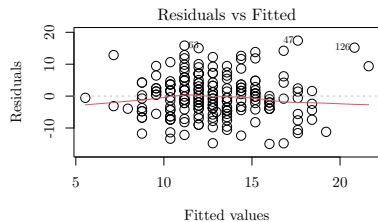


Diagnostics

```
plot( m1 )
```

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```
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```



Evaluation of estimates

```
summary( m1 )
```

Evaluation of estimates

```
summary( m1 )
```

Call:

```
lm(formula = ChildBeh ~ ParSty, data = ESdata)
```

Residuals:

Min	1Q	Median	3Q	Max
-14.994	-4.038	-0.277	3.636	17.397

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.928	1.601	2.45	0.015
ParSty	0.804	0.139	5.79	2.8e-08

Residual standard error: 6.47 on 198 degrees of freedom

Multiple R-squared: 0.145, Adjusted R-squared: 0.14

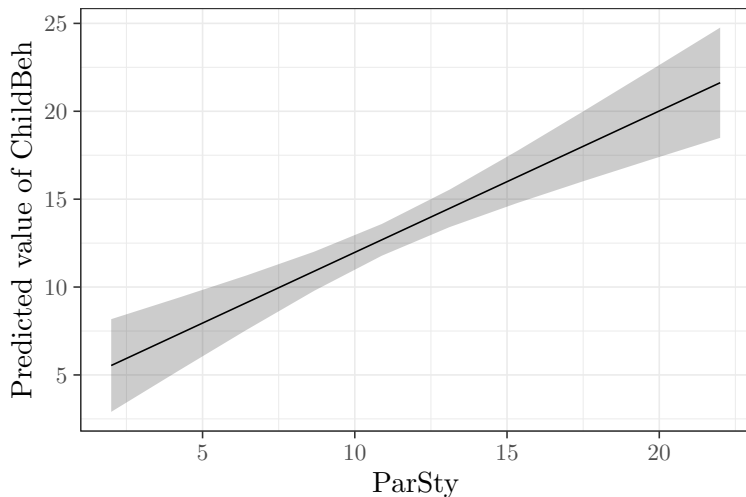
F-statistic: 33.5 on 1 and 198 DF, p-value: 2.75e-08

Graphical representation

```
library( modelbased )  
plot( estimate_relation( m1 ) )
```

Graphical representation

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library( modelbased )  
plot( estimate_relation( m1 ) )
```



One dichotomic predictor

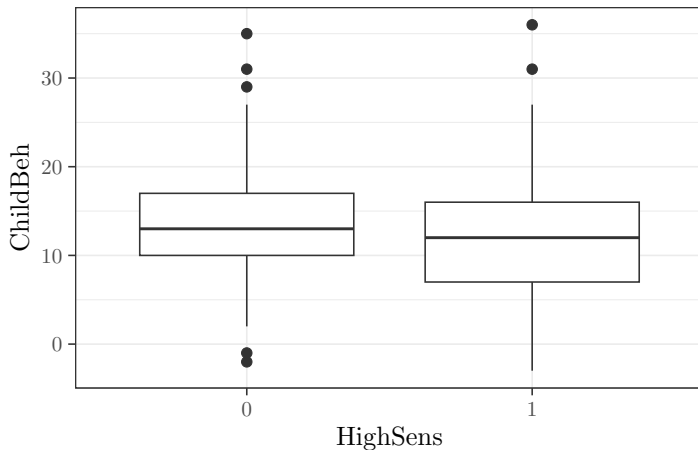
We want to assess whether the type of sensitivity is predictive of behavioural problems.

One dichotomic predictor

```
plot( ChildBeh ~ HighSens, data = ESdata )
```

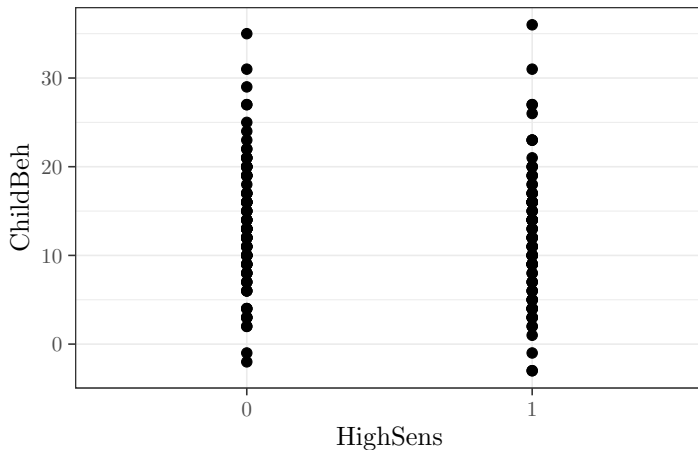
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One dichotomic predictor

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One dichotomic predictor

```
m2 <- lm( ChildBeh ~ HighSens, data = ESdata )
```

One dichotomic predictor

```
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```

```
coef( m2 )  
(Intercept)    HighSens1  
    13.53333     -1.52281
```

The regression equation will be

One dichotomic predictor

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```

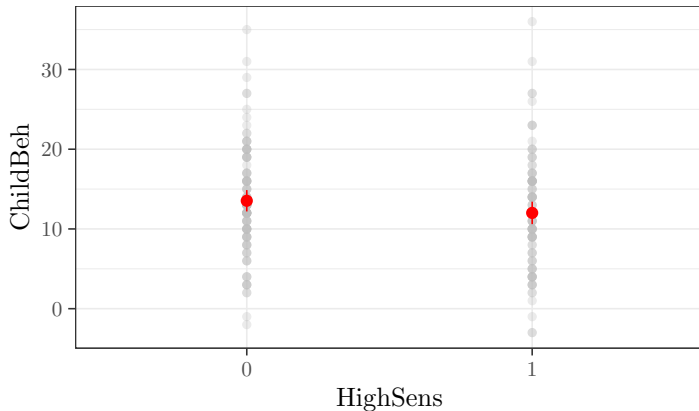
```
coef( m2 )  
(Intercept)    HighSens1  
    13.53333    -1.52281
```

The regression equation will be

$$\hat{Y} = 13.53 - 1.52D$$

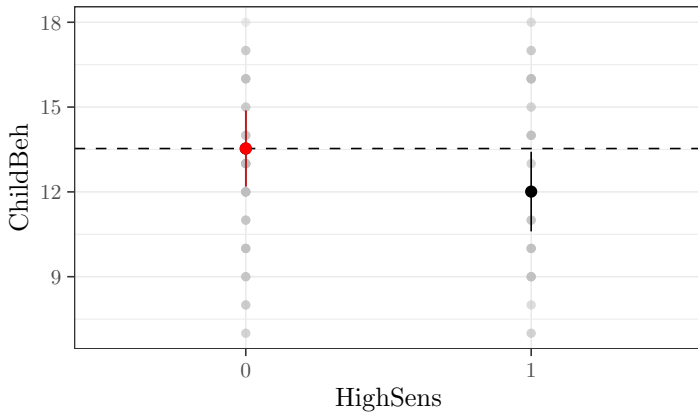
Interpretation of parameters

- The points on the regression line $\hat{Y}$ indicate the expected values of **ChildBeh** in relation to the Sensitivity level (X).



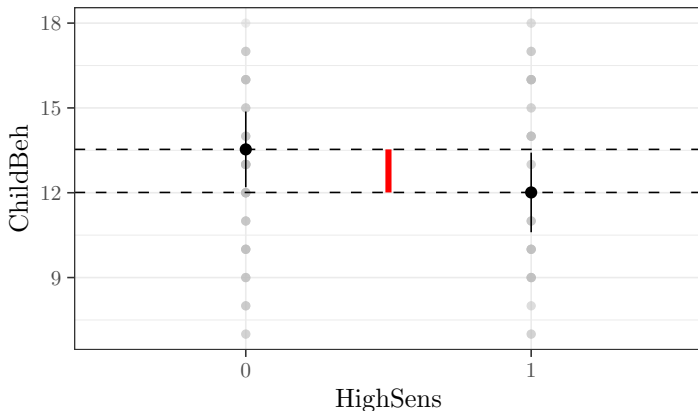
Interpretation of parameters

- The intercept $b_0 = 13.53$ indicates the expected score for ChildBeh for a subject with HighSens equal to zero.



Interpretation of parameters

- The regression coefficient $b_1 = -1.52$ indicates that, when looking at the **ChildBeh** scores in a subpopulation where the **HighSens** values differ by one unit, we would expect an average difference of -1.52 points.

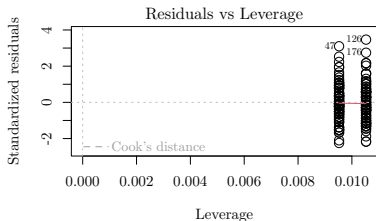
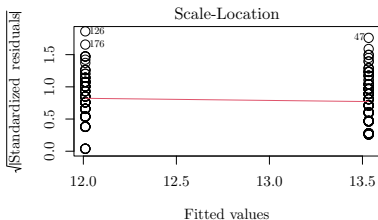
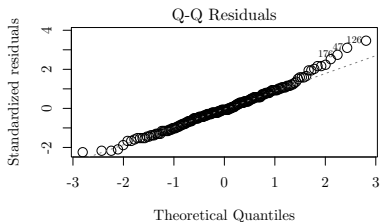
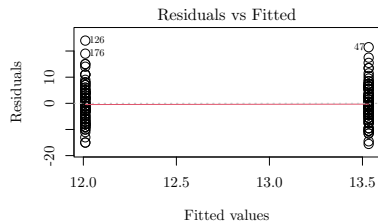


Diagnostics

```
plot( m2 )
```

Diagnostics

```
plot( m2 )
```



Evaluation of estimates

```
summary( m2 )
```

Call:

```
lm(formula = ChildBeh ~ HighSens, data = ESdata)
```

Residuals:

Min	1Q	Median	3Q	Max
-15.533	-4.533	-0.533	3.989	23.989

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.533	0.679	19.93	<2e-16
HighSens1	-1.523	0.985	-1.55	0.12

Residual standard error: 6.96 on 198 degrees of freedom

Multiple R-squared: 0.0119, Adjusted R-squared: 0.00693

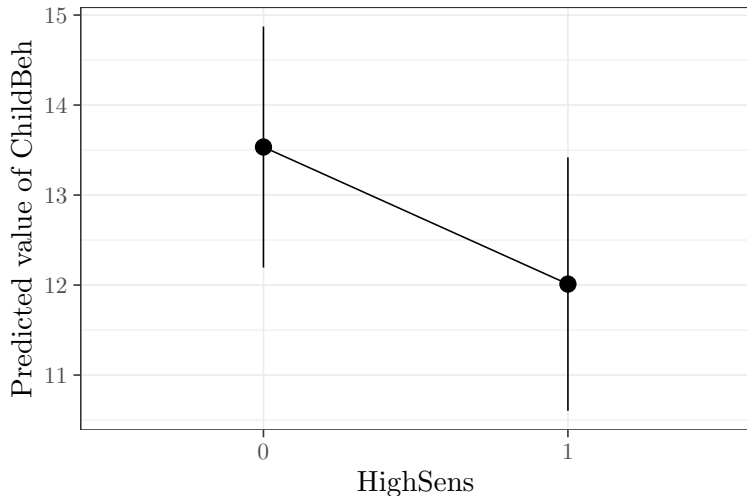
F-statistic: 2.39 on 1 and 198 DF, p-value: 0.124

Graphical representation

```
plot( estimate_relation( m2 ) )
```

Graphical representation

```
plot( estimate_relation( m2 ) )
```



The model just presented is known as

**The model just presented is known as
the Student's t-test**

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```
t.test( ChildBeh ~ HighSens, data = ESdata,  
        var.equal = TRUE)
```

```
^^ITwo Sample t-test
```

```
data:  ChildBeh by HighSens
```

```
t = 1.55, df = 198, p-value = 0.12
```

```
sample estimates:
```

mean in group 0	mean in group 1
13.533	12.011

The model just presented is known as the Student's t-test

```
summary( lm( ChildBeh ~ HighSens, data = ESdata ) )
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	13.533	0.679	19.93	<2e-16
HighSens1	-1.523	0.985	-1.55	0.12

Residual standard error: 7 on 198 degrees of freedom

The model just presented is known as the Student's t-test or the Analysis of Variance

```
anova( lm( ChildBeh ~ HighSens, data = ESdata ) )
```

Analysis of Variance Table

Response: ChildBeh

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
HighSens	1	116	115.7	2.39	0.12
Residuals	198	9585	48.4		

The model just presented is known as the Student's t-test or the Analysis of Variance

```
anova( lm( ChildBeh ~ HighSens, data = ESdata ) )
```

Analysis of Variance Table

Response: ChildBeh

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
HighSens	1	116	115.7	2.39	0.12
Residuals	198	9585	48.4		

We see that the p -value is the same, and we note that $F = 2.39$ is the square of $t = 1.55$:

$$t^2 = 1.55^2 = 2.39$$

Student's t -test is the equivalent of a linear model in which the dependent variable is a quantitative variable and the predictor is a dichotomous variable.

If we know how to perform linear regression, we also know how to perform the t -test, and vice versa.

Multiple linear model

Multiple regression

When the model has more predictors, the formulation may be as follows:

1

$$\mathbf{Y} = \mathbf{X}\beta + \epsilon$$

in matrix form.

Multiple regression

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$$\mathbf{Y} = \mathbf{X}\beta + \epsilon$$

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2

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon$$

in extended form.

Multiple regression

When the model has more predictors, the formulation may be as follows:

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$$\mathbf{Y} = \mathbf{X}\beta + \epsilon$$

in matrix form.

2

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \epsilon$$

in extended form.

3

When the model has several predictors (quantitative and/or categorical) and a single (quantitative) dependent variable Y , it is called **Multiple Linear Regression**.

Let us consider a model for predicting behavioural problems based on parenting style and sensitivity.

Additive model

```
m3 <- lm( ChildBeh ~ ParSty + HighSens, data = ESdata )  
coef( m3 )
```

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(Intercept)	ParSty	HighSens1
4.6	0.8	-1.3

Additive model

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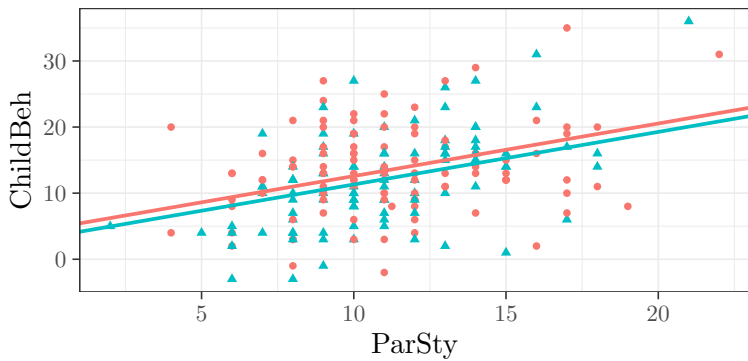
$$\left\{ \begin{array}{l} \hat{Y} = 4.63 + 0.8X - 1.27 \times 0 \\ \quad = 4.63 + 0.8X \end{array} \right. \quad \text{if } D = 0$$

Additive model

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m3 <- lm( ChildBeh ~ ParSty + HighSens, data = ESdata )  
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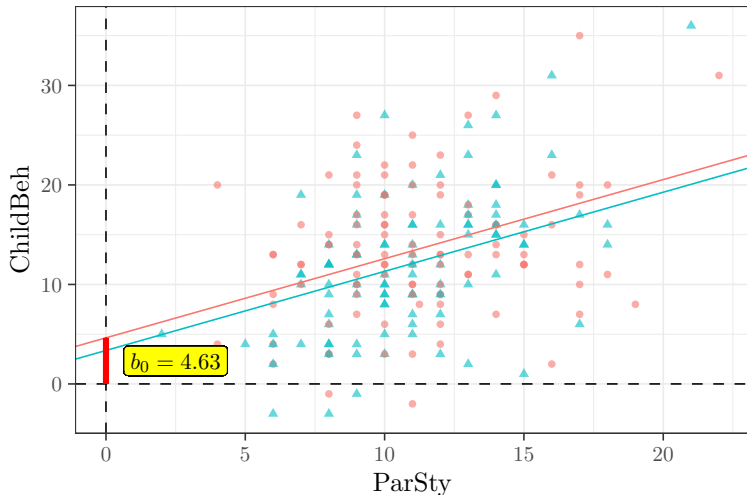
$$\left\{ \begin{array}{ll} \hat{Y} = 4.63 + 0.8X - 1.27 \times 0 \\ \quad = 4.63 + 0.8X & \text{if } D = 0 \\ \hat{Y} = 4.63 + 0.8X - 1.27 \times 1 \\ \quad = 3.37 + 0.8X & \text{if } D = 1 \end{array} \right.$$



HighSens ● 0 ▲ 1

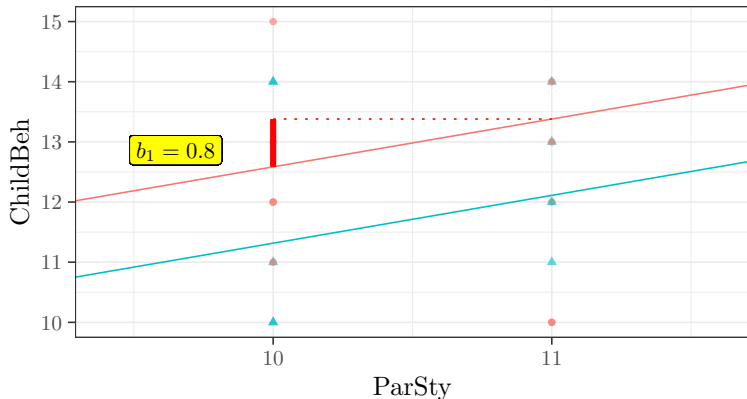
Interpretation of parameters

- $b_0 = 4.63$ (intercept): is the expected ChildBeh score for a subject with low sensitivity and a ParSty score of 0.



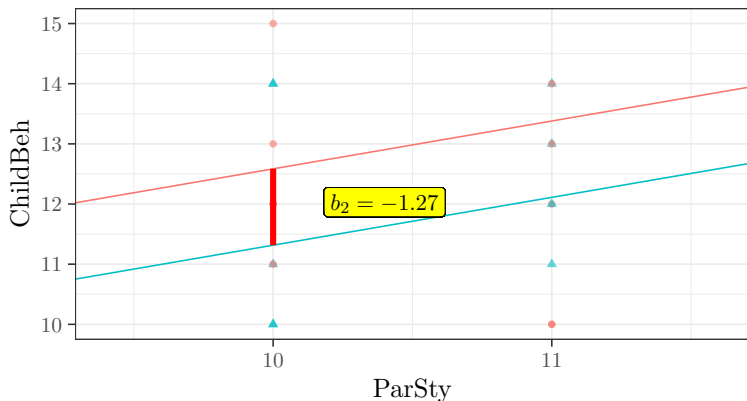
Interpretation of parameters

- $b_1 = 0.8$ (coefficient relating to ParSty) is the expected difference in behavioural problems between two subjects, with the same sensitivity type, who differ by 1 point on parenting style.



Interpretation of parameters

- $b_2 = -1.27$ (coefficient relating to HighSens) is the expected difference in behavioural problems between two subjects, with the same level of ParSty, who differ in their type of sensitivity.



Analysis of estimates

```
summary( m3 )
```

Call:

```
lm(formula = ChildBeh ~ ParSty + HighSens, data = ESdata)
```

Residuals:

Min	1Q	Median	3Q	Max
-15.380	-4.012	-0.348	3.595	16.849

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.634	1.677	2.76	0.0063
ParSty	0.795	0.139	5.73	3.8e-08
HighSens1	-1.269	0.916	-1.39	0.1673

Residual standard error: 6.5 on 197 degrees of freedom

Multiple R-squared: 0.153, Adjusted R-squared: 0.144

F-statistic: 17.8 on 2 and 197 DF, p-value: 7.89e-08

Predictive check

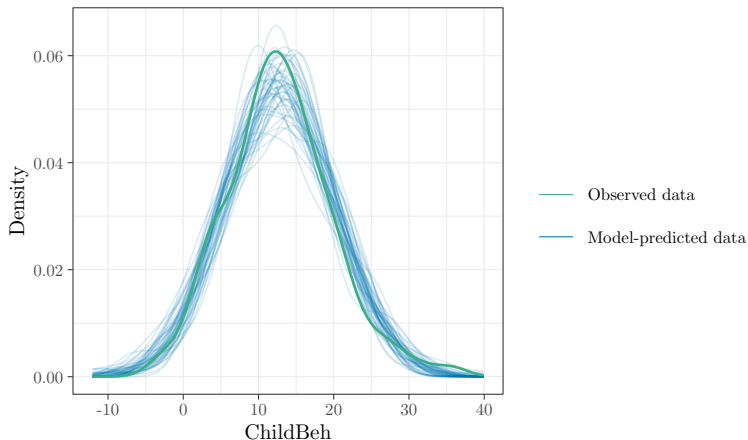
```
library( performance )  
check_predictions( m3 )
```

Predictive check

```
library( performance )  
check_predictions( m3 )
```

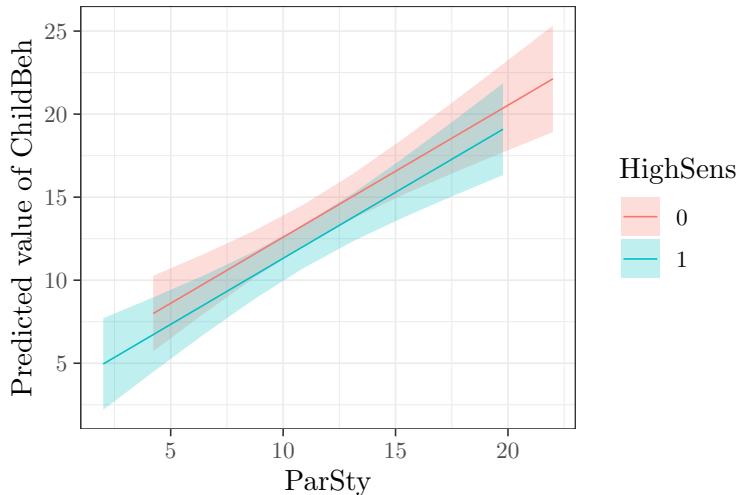
Posterior Predictive Check

Model-predicted lines should resemble observed data line



Model predictions

```
plot( estimate_relation( m3 ) )
```



Interaction model

Let us consider a model for predicting problematic behaviour based on ParSty and HighSens, whilst also including the interaction between the predictors.

Interaction model

```
m4 <- lm( ChildBeh ~ ParSty * HighSens, data = ESdata )  
coef( m4 )
```

Interaction model

```
m4 <- lm( ChildBeh ~ ParSty * HighSens, data = ESdata )  
coef( m4 )
```

(Intercept)	ParSty	HighSens1
8.14	0.48	-8.99
ParSty:HighSens1		
0.70		

Interaction model

```
m4 <- lm( ChildBeh ~ ParSty * HighSens, data = ESdata )  
coef( m4 )
```

(Intercept)	ParSty	HighSens1
8.14	0.48	-8.99
ParSty:HighSens1		
0.70		

$$\left\{ \begin{array}{l} \hat{Y} = 8.14 + 0.48X - 8.99D + 0.7XD \\ \quad = 8.14 + 0.48X \end{array} \right. \quad \text{if } D = 0$$

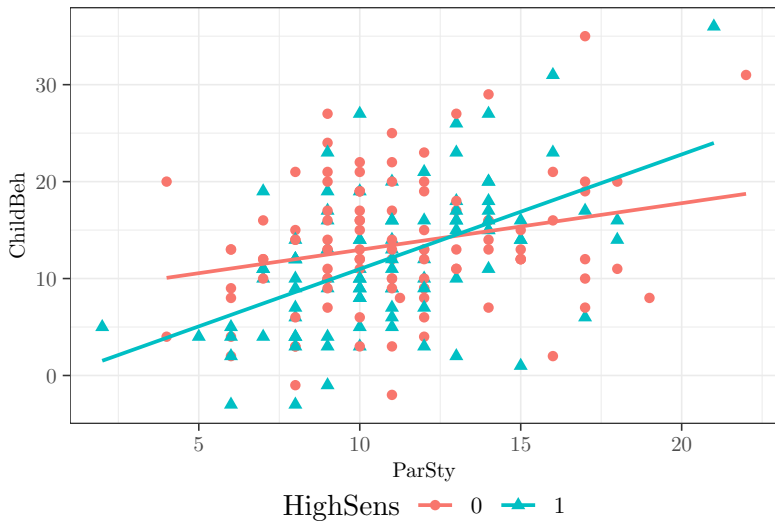
Interaction model

```
m4 <- lm( ChildBeh ~ ParSty * HighSens, data = ESdata )  
coef( m4 )
```

(Intercept)	ParSty	HighSens1
8.14	0.48	-8.99
ParSty:HighSens1		
0.70		

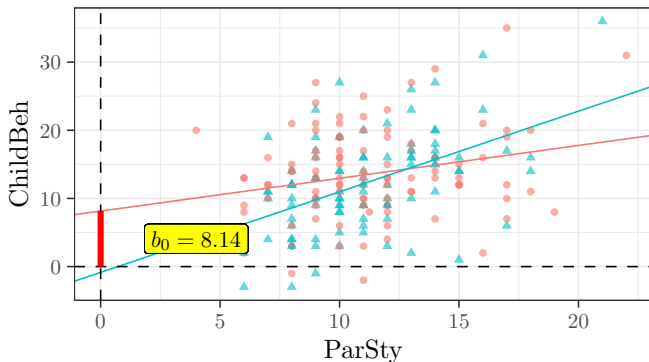
$$\left\{ \begin{array}{ll} \hat{Y} = 8.14 + 0.48X - 8.99D + 0.7XD \\ \quad = 8.14 + 0.48X & \text{if } D = 0 \\ \hat{Y} = 8.14 + 0.48X - 8.99D + 0.7XD \\ \quad = -0.85 + 1.18X & \text{if } D = 1 \end{array} \right.$$

Interaction model



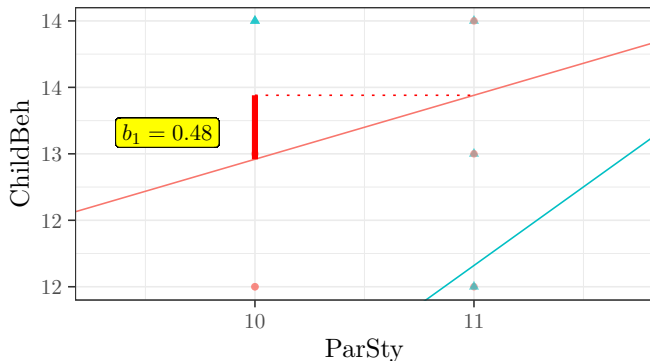
Interpretation of parameters

- $b_0 = 8.14$ (intercept): expected ChildBeh score for a subject with low sensitivity score and ParSty score = 0.



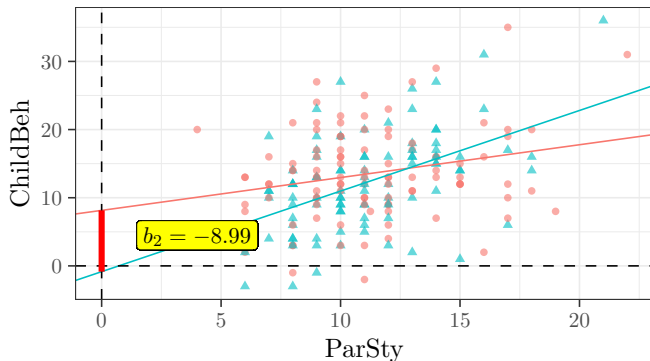
Interpretation of parameters

- $b_1 = 0.48$ (ParSty coefficient): expected score difference between subjects with low sensitivity, who differ by one point in the ParSty scores.



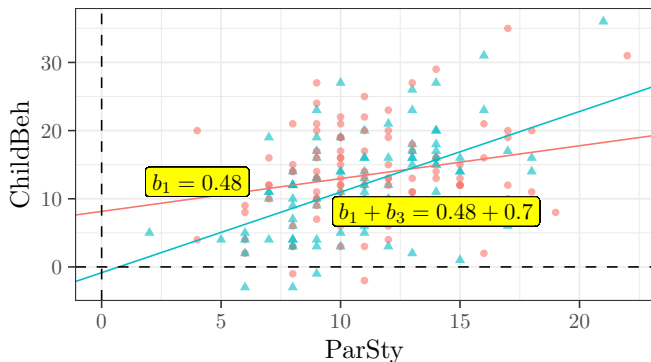
Interpretation of parameters

- $b_2 = -8.99$ (HighSens coefficient) expected difference in behavioural problems between subjects with $\text{ParSty} = 0$ and different types of sensitivity.



Interpretation of parameters

- $b_3 = 0.7$ (interaction coefficient) difference in the slopes of the regression lines as a function of **ParSty**, comparing subjects with different types of sensitivity.



Analysis of estimates

```
summary( m4 )
```

Call:

```
lm(formula = ChildBeh ~ ParSty * HighSens, data = ESdata)
```

Residuals:

Min	1Q	Median	3Q	Max
-15.889	-4.003	-0.574	3.532	18.671

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.144	2.153	3.78	0.00021
ParSty	0.482	0.184	2.61	0.00964
HighSens1	-8.990	3.165	-2.84	0.00498
ParSty:HighSens1	0.701	0.275	2.55	0.01169

Residual standard error: 6.4 on 196 degrees of freedom

Multiple R-squared: 0.18, Adjusted R-squared: 0.168

F-statistic: 14.3 on 3 and 196 DF, p-value: 1.72e-08

Predictive check

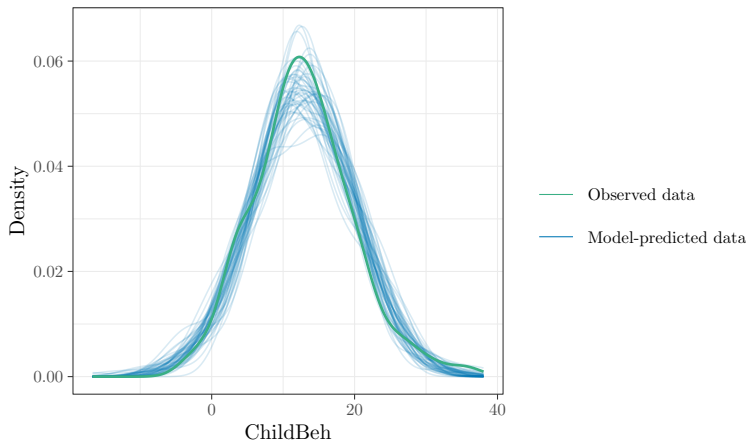
```
check_predictions( m4 )
```

Predictive check

```
check_predictions( m4 )
```

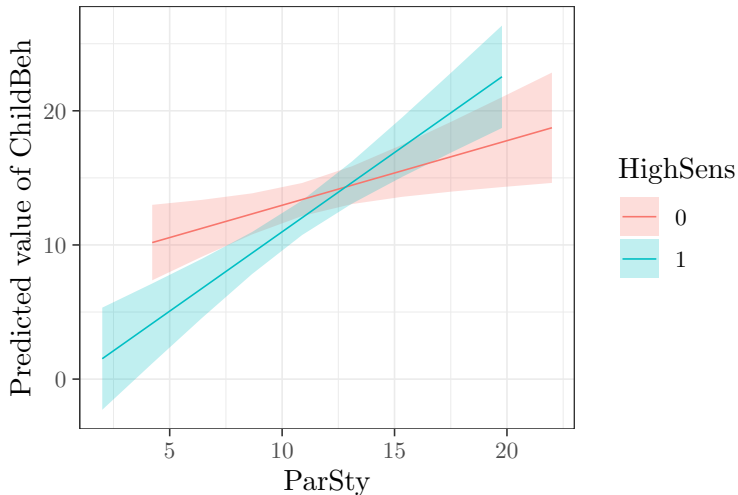
Posterior Predictive Check

Model-predicted lines should resemble observed data line



Model predictions

```
plot( estimate_relation( m4 ) )
```



Effect sizes

- With R^2 :

Effect sizes

- With R^2 :

```
summary( m4 )$r.squared
```

Effect sizes

- With R^2 :

```
summary( m4 )$r.squared  
[1] 0.18
```


Effect sizes

- With R^2 :

```
summary( m4 )$r.squared  
[1] 0.18
```

- With η^2 :

Effect sizes

- With R^2 :

```
summary( m4 )$r.squared  
[1] 0.18
```

- With η^2 :

```
library( heplots )  
etasq( m4, partial = FALSE )
```

Effect sizes

- With R^2 :

```
summary( m4 )$r.squared  
[1] 0.18
```

- With η^2 :

```
library( heplots )  
etasq( m4, partial = FALSE )  
  
              eta^2  
ParSty        0.1411  
HighSens      0.0083  
ParSty:HighSens 0.0271  
Residuals      NA
```

R^2 and η^2 are closely related.

- They measure the proportion of explained variability with respect to the total variability.
- R^2 is unique, it expresses the proportion of variability relative to all predictors.
- The η^2 values are as many as the number of predictors and measure the proportion of variability explained by each.
- With only one predictor, $R^2 = \eta^2$

Interactions

Tests of Interactions

In the context of differential susceptibility, there are different tests, such as:

- 1 the **regions of significance** (RoS, Roisman et al., 2012)
- 2 the **proportion of interaction** (PoI)
- 3 the **proportion of affected** (PA, Roisman et al., 2012).

An on-line resource:

<https://yourpersonality.net/interaction/>

Del Giudice, M. (2017). Statistical tests of differential susceptibility: Performance, limitations, and improvements. *Development and Psychopathology*, 29, 1267–1278.

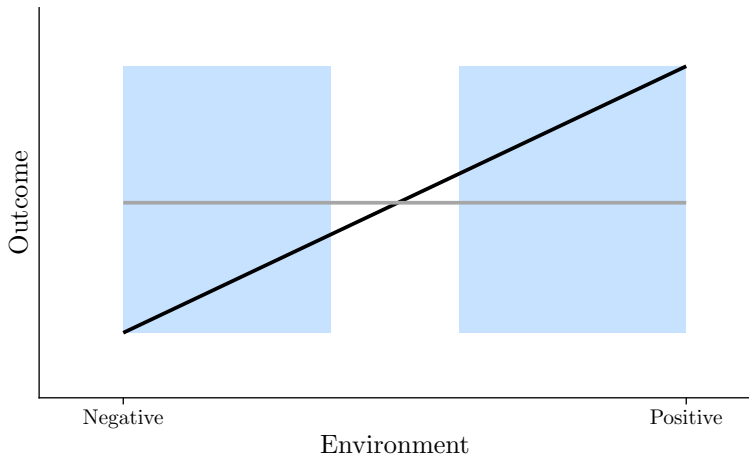
Roisman, G. I., Newman, D. A., Fraley, C., Haltigan, J. D., Groh, A. M., & Haydon, K. C. (2012). Distinguishing differential susceptibility from diathesis-stress: Recommendations for evaluating interaction effects. *Development and Psychopathology*, 24, 389–409.

Regions of significance

- RoS delimit values of the environmental variable for which the moderator is significantly associated with the outcome variable (Kochanska, Kim, Barry, & Philibert, 2011).
- A significant interaction is deemed consistent with differential susceptibility if RoS extend to both the low and high end of a conventional interval of ± 2 SD from the mean of the environmental variable.

Kochanska, G., Kim, S., Barry, R. A., & Philibert, R. A. (2011). Children's genotypes interact with maternal responsive care in predicting children's competence: Diathesis-stress or differential susceptibility? *Development and Psychopathology*, 23, 605–616.

RoS with respect to X

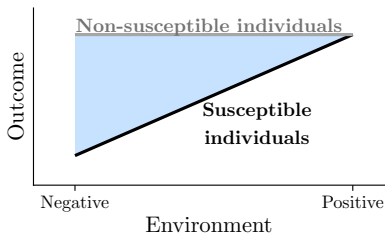


Proportion of interaction

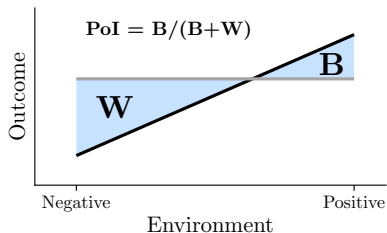
- The PoI is the proportion of the total area between the lines of an interaction plot, bounded by ± 2 SD of the environmental variable, that lies on the positive side of the crossover point, where “positive” refers to the quality of the environment.
- It is obtained by dividing the amount of change “for better” (B) by its sum with the amount of change “for worse” (W): $\text{PoI} = B/(B + W)$.

[A] PoI = .00

Diathesis-stress

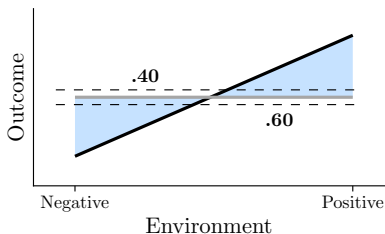


[B] PoI = .25



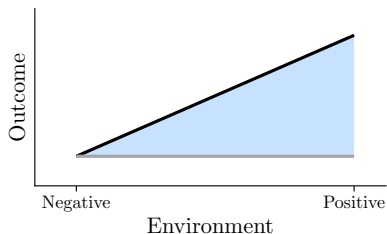
[C] PoI = .50

Differential susceptibility



[D] PoI = 1.00

Vantage sensitivity



Proportion of affected

- The PA is the proportion of individuals in the sample who fall on the positive side of the crossover point.
- Based on the assumption of approximate normality in the environmental variable, interactions are deemed consistent with differential susceptibility if $PA > .16$.
- Most researchers have adopted it in conjunction with the .40 – .60 window for PoI as a criterion for screening potential differential susceptibility interactions.

PoI.PA() function

```
PoI.PA <- function( Z, yvar = Y, xvar = X, dvar = D ){  
  
  fit <- lm( formula( paste0(yvar, " ~ ", xvar, "*", dvar) )  
    , data = Z )  
  BE <- coef(fit)  
  int.point <- BE[3]*(-1) / BE[4]  
  PA <- sum( fit$model[,xvar] > int.point ) /  
    length(fit$model[,xvar])  
  
  PoI <- ( 1 + ( BE[4]*min(Z[,xvar]) + BE[3] )^2 /  
    ( BE[4]*max(Z[,xvar]) + BE[3] )^2 ) ^ (-1)  
  
  return( list( PA = PA, PoI = as.numeric(PoI),  
    int.point = as.numeric(int.point) ))  
}
```

Analyze our data

- The model (m4):

```
ChildBeh ~ ParSty * HighSens
```

Analyze our data

- The model (m4):

```
ChildBeh ~ ParSty * HighSens
```

- Regions of Significance:

```
library( reghelper )  
sig_regions( m4 )
```

Analyze our data

- The model (m4):

```
ChildBeh ~ ParSty * HighSens
```

- Regions of Significance:

```
library( reghelper )  
sig_regions( m4 )
```

```
lower upper  
10      21
```

Analyze our data

- The model (m4):

```
ChildBeh ~ ParSty * HighSens
```

- Regions of Significance:

```
library( reghelper )  
sig_regions( m4 )
```

```
lower upper  
10      21
```

- PoI and PA:

```
PoI.PA( ESdata,  
  yvar = "ChildBeh", xvar = "ParSty", dvar = "HighSens" )
```


Analyze our data

- The model (m4):

```
ChildBeh ~ ParSty * HighSens
```

- Regions of Significance:

```
library( reghelper )  
sig_regions( m4 )
```

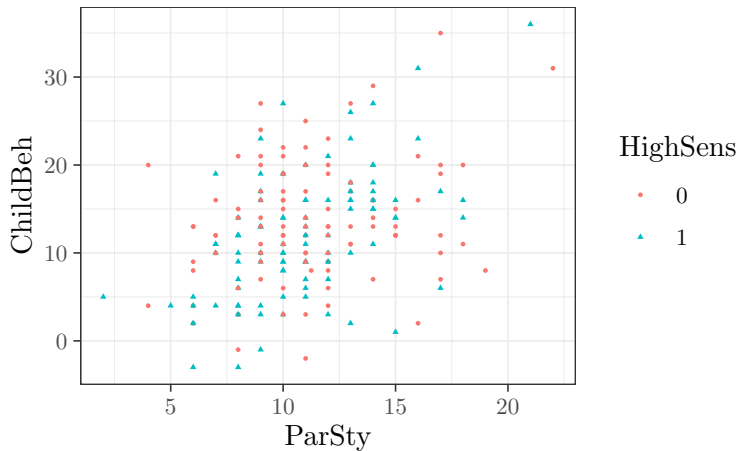
```
lower upper  
10      21
```

- PoI and PA:

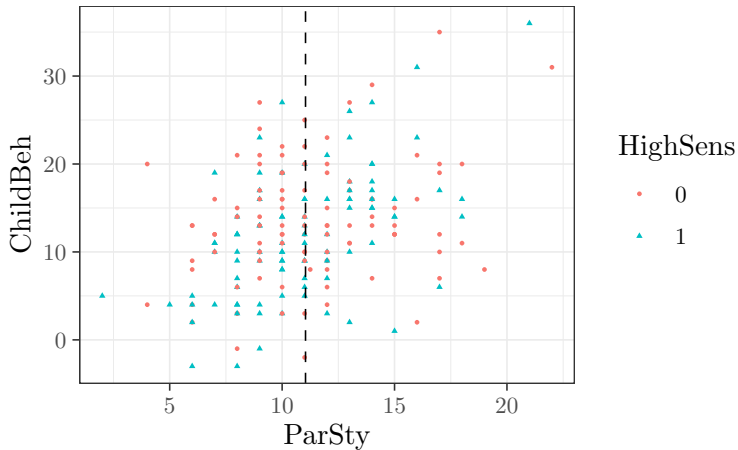
```
PoI.PA( ESdata,  
  yvar = "ChildBeh", xvar = "ParSty", dvar = "HighSens" )
```

```
$PoI  
[1] 0.42  
$PA  
[1] 0.29
```

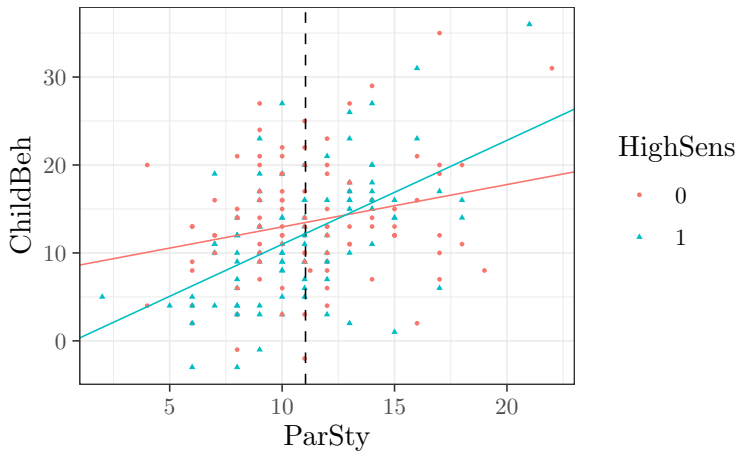
The data



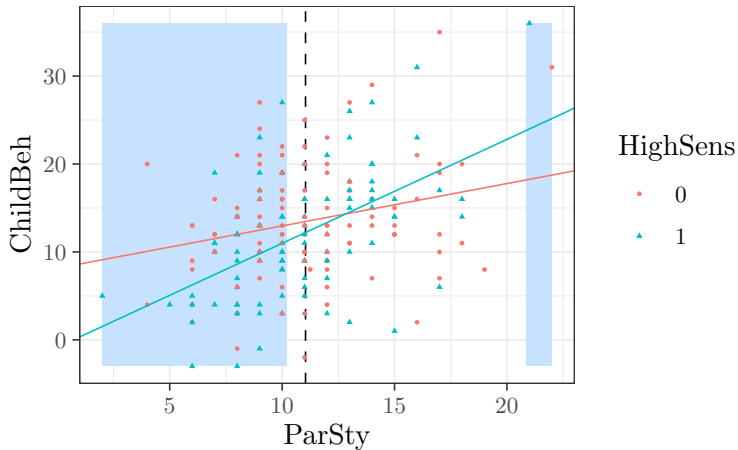
Mean score of X



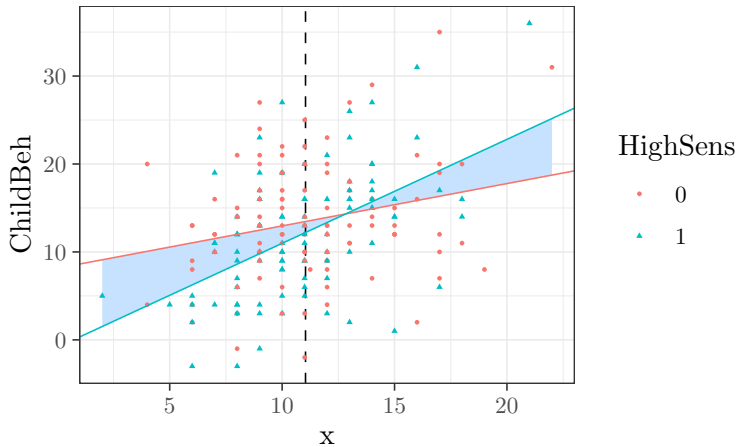
Regression lines



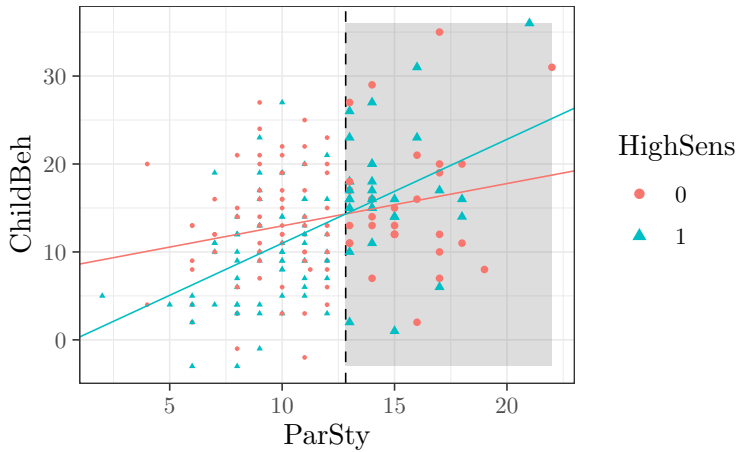
$$\text{RoS} = [10.2, 20.9]$$



PoI = 0.42



PA = 0.3



Lionetti, F., Calcagnì, A., D'Urso, G., Spinelli, M., Fasolo, M., Pluess, M., Pastore M. (2024). A Bayesian approach for exploring person $\times$ environment interaction within the environmental sensitivity meta-framework. *Journal of Child Psychology and Psychiatry*,
<https://doi.org/10.1111/jcpp.14000>

Models comparison and models selection

*All models are wrong,
but some are useful.*
(George E.P. Box)

- The aim of a (statistical) model is essentially the following: starting from a set of observed data, identify the underlying latent process that generated these data.
- The goal of a researcher is to identify the *best model*.

Model comparison

- A statistical model is a mathematical formalization of relationships among variables.
- Many statistical tests can be described in the form of a statistical model (e.g., t test).
- The different statistical models on the same dependent variable can be compared to find the one that best explains the data.
- The model comparison can be performed in various ways: with ANOVA, with Information Criteria (e.g., AIC, BIC), with the Bayes Factor and so on.

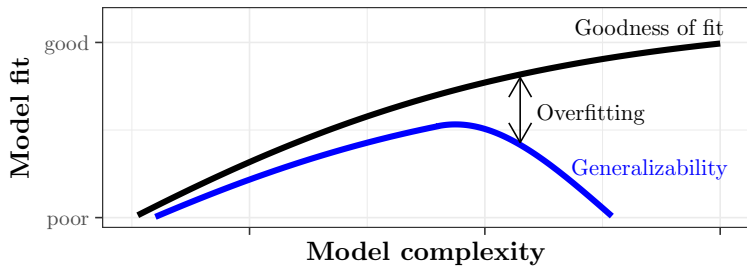
Problems

- There can be many models that explain the same data.
- Random variations can make it difficult to identify the model.
- The true model may not be among those actually tested.

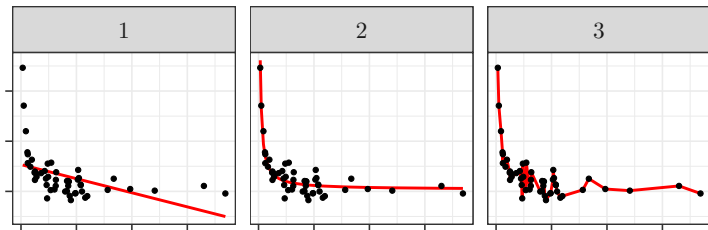
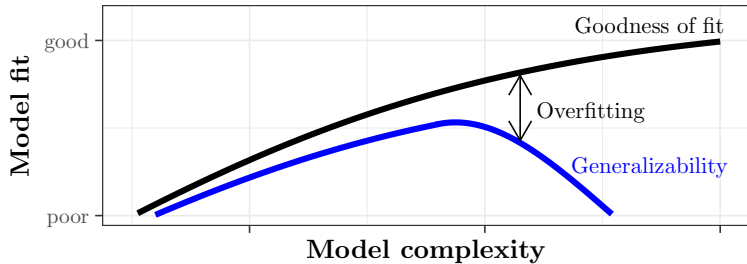
Problems

- There can be many models that explain the same data.
- Random variations can make it difficult to identify the model.
- The true model may not be among those actually tested.

Therefore, a realistic objective is to identify the model representing the best approximation to the true model.

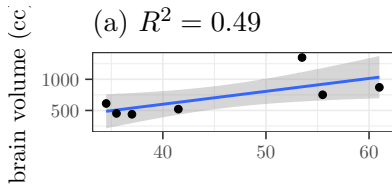


Pitt, Myung & Zhang, 2002



Pitt, Myung & Zhang, 2002

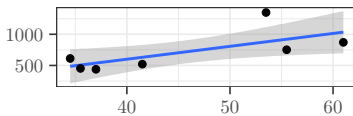
... *fitting is easy; prediction is hard*
(Richard McElreath)



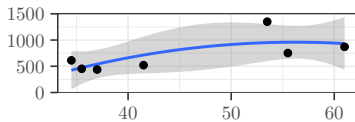
... *fitting is easy; prediction is hard*
(Richard McElreath)

brain volume (cc)

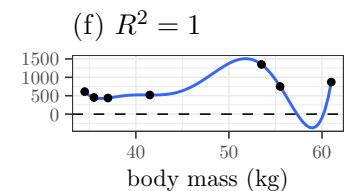
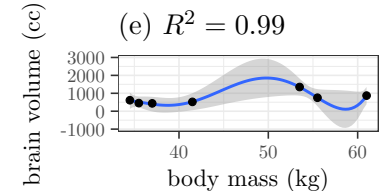
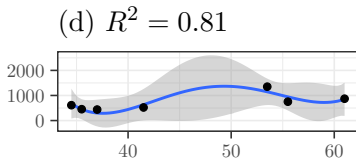
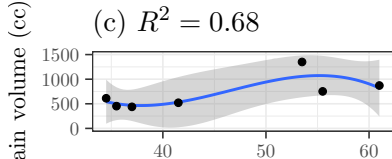
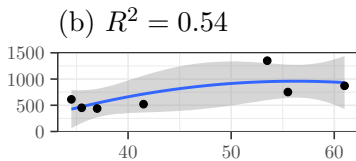
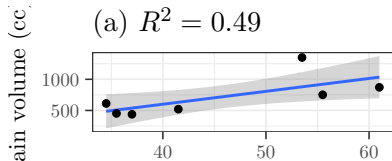
(a) $R^2 = 0.49$



(b) $R^2 = 0.54$



... fitting is easy; prediction is hard
(Richard McElreath)



Model comparison using ANOVA

```
m0 <- lm( ChildBeh ~ 1, data = ESdata )  
m1 <- lm( ChildBeh ~ ParSty, data = ESdata )  
m2 <- lm( ChildBeh ~ HighSens, data = ESdata )  
m3 <- lm( ChildBeh ~ ParSty + HighSens, data = ESdata )  
m4 <- lm( ChildBeh ~ ParSty * HighSens, data = ESdata )  
anova( m0, m1, m2, m3, m4 )
```

Model comparison using ANOVA

```
m0 <- lm( ChildBeh ~ 1, data = ESdata )
m1 <- lm( ChildBeh ~ ParSty, data = ESdata )
m2 <- lm( ChildBeh ~ HighSens, data = ESdata )
m3 <- lm( ChildBeh ~ ParSty + HighSens, data = ESdata )
m4 <- lm( ChildBeh ~ ParSty * HighSens, data = ESdata )
anova( m0, m1, m2, m3, m4 )
```

Analysis of Variance Table

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	199	9701				
2	198	8297	1	1404	34.60	1.7e-08
3	198	9585	0	-1288		
4	197	8217	1	1368	33.72	2.5e-08
5	196	7954	1	263	6.48	0.012

Model comparison: AIC and BIC

- Two widely used indices for model comparisons are AIC (Akaike Information Criterion; Akaike, 1974) and BIC (Bayesian Information Criterion; Schwarz, 1978).
- Such indices measure the efficiency of the model in terms of data prediction.
- Given k AIC/BIC for k different models, the model with the lowest AIC/BIC value is preferred.

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- Given k AIC/BIC for k different models, the model with the lowest AIC/BIC value is preferred.

```
A <- AIC(m0, m1, m2, m3, m4)
```

```
A[ order( A[,2] ), ]
```

	df	AIC
m4	5	1314
m1	3	1319
m3	4	1319
m2	3	1348
m0	2	1348

Model comparison: AIC and BIC

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```
A <- AIC(m0, m1, m2, m3, m4)
```

```
A[ order( A[,2] ), ]
```

	df	AIC
m4	5	1314
m1	3	1319
m3	4	1319
m2	3	1348
m0	2	1348

```
B <- BIC(m0, m1, m2, m3, m4)
```

```
B[ order( B[,2] ), ]
```

	df	BIC
m1	3	1329
m4	5	1331
m3	4	1332
m0	2	1355
m2	3	1357

Quantification of evidence (approximate method)

- A big advantage in the use of AIC/BIC is the possibility to estimate the relative plausibility of one model compared to another.
- We can establish how much one model is more plausible than another by avoiding all problems associated with p -value.
- In addition, the computation of relative evidence (or evidence ratio) is very simple:
 - 1 Let Δ be the difference between the AIC/BIC values of the two models
 - 2 Calculate $\ell = \exp(\Delta/2) = e^{\frac{\Delta}{2}}$

Quantification of evidence (approximate method)

Suppose you want to compare the model with interaction (m4) with the one without (m3)

Quantification of evidence (approximate method)

Suppose you want to compare the model with interaction (m4)
with the one without (m3)

```
(DAIC <- AIC(m3) - AIC(m4))
```

```
exp( DAIC / 2 )
```

Quantification of evidence (approximate method)

Suppose you want to compare the model with interaction (m4)
with the one without (m3)

```
(DAIC <- AIC(m3) - AIC(m4))
```

```
[1] 4.5
```

```
exp( DAIC / 2 )
```

```
[1] 9.5
```

Conclusion with AIC:

Quantification of evidence (approximate method)

Suppose you want to compare the model with interaction (m4) with the one without (m3)

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Conclusion with AIC:
the model with interaction
is about 10 times more
plausible than the other
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Quantification of evidence (approximate method)

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```
(DAIC <- AIC(m3) - AIC(m4))
```

```
[1] 4.5
```

```
exp( DAIC / 2 )
```

```
[1] 9.5
```

```
(DBIC <- BIC(m3) - BIC(m4))
```

```
exp( DBIC / 2 )
```

Conclusion with AIC:
the model with interaction
is about 10 times more
plausible than the other
(without interaction).

Quantification of evidence (approximate method)

Suppose you want to compare the model with interaction (m4)
with the one without (m3)

```
(DAIC <- AIC(m3) - AIC(m4))
```

```
[1] 4.5
```

```
exp( DAIC / 2 )
```

```
[1] 9.5
```

Conclusion with AIC:
the model with interaction
is about 10 times more
plausible than the other
(without interaction).

```
(DBIC <- BIC(m3) - BIC(m4))
```

```
[1] 1.2
```

```
exp( DBIC / 2 )
```

```
[1] 1.8
```

Conclusion with BIC:

Quantification of evidence (approximate method)

Suppose you want to compare the model with interaction (m4) with the one without (m3)

```
(DAIC <- AIC(m3) - AIC(m4))
```

```
[1] 4.5
```

```
exp( DAIC / 2 )
```

```
[1] 9.5
```

Conclusion with AIC:
the model with interaction
is about 10 times more
plausible than the other
(without interaction).

```
(DBIC <- BIC(m3) - BIC(m4))
```

```
[1] 1.2
```

```
exp( DBIC / 2 )
```

```
[1] 1.8
```

Conclusion with BIC:
the model with interaction
is about 2 times more
plausible than the other
(without interaction).

Relative likelihood

```
A$delta <- max( A$AIC ) - A$AIC  
A$rl <- exp( A$delta / 2 )  
A[ order(A$AIC), ]
```


Relative likelihood

```
A$delta <- max( A$AIC ) - A$AIC  
A$r1 <- exp( A$delta / 2)  
A[ order(A$AIC), ]
```

	df	AIC	delta	r1
m4	5	1314	33.7	2.1e+07
m1	3	1319	29.3	2.3e+06
m3	4	1319	29.2	2.2e+06
m2	3	1348	0.4	1.2e+00
m0	2	1348	0.0	1.0e+00

Relative likelihood

```
A$delta <- max( A$AIC ) - A$AIC  
A$rl <- exp( A$delta / 2)  
A[ order(A$AIC), ]
```

	df	AIC	delta	rl
m4	5	1314	33.7	2.1e+07
m1	3	1319	29.3	2.3e+06
m3	4	1319	29.2	2.2e+06
m2	3	1348	0.4	1.2e+00
m0	2	1348	0.0	1.0e+00

The model m4 ($\text{ChildBeh} \sim \text{ParSty} * \text{HighSens}$) turns out to be about $2.09\text{e}+07$ times more likely than the null model.

Akaike weights

Finally, we can normalize the relative model likelihoods to obtain Akaike weights, i.e. the probability of each model M_i , given the data and the set of considered models ($\mathcal{M}$):

$$w_i = \text{Prob}(\text{model } M_i | \text{data}, \mathcal{M}) = \frac{\ell_i}{\sum_{j=1}^J \ell_j}$$

where J is the number of considered models; in our case $J = 5$.

Akaike weights

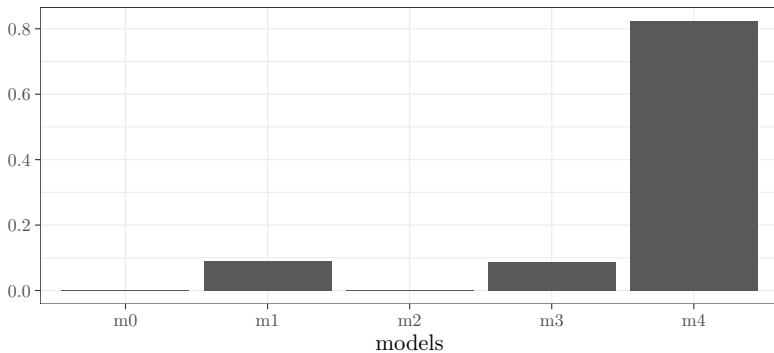
Finally, we can normalize the relative model likelihoods to obtain Akaike weights, i.e. the probability of each model M_i , given the data and the set of considered models ($\mathcal{M}$):

$$w_i = \text{Prob}(\text{model } M_i | \text{data}, \mathcal{M}) = \frac{\ell_i}{\sum_{j=1}^J \ell_j}$$

where J is the number of considered models; in our case $J = 5$.

```
A$w <- A$r1 / sum( A$r1 )
```

```
barplot( A$w, names.arg = rownames(A), xlab = "models" )
```



The AICcmodavg package

```
library( AICcmodavg )  
Models <- list( m0=m0, m1=m1, m2=m2, m3=m3, m4=m4 )  
(AICtable <- aictab( Models ))
```

The AICcmodavg package

```
library( AICcmodavg )  
Models <- list( m0=m0, m1=m1, m2=m2, m3=m3, m4=m4 )  
(AICtable <- aictab( Models ))
```

Model selection based on AICc:

	K	AICc	Delta_AICc	AICcWt	Cum.Wt	LL
m4	5	1314	0.0	0.81	0.81	-652
m1	3	1319	4.3	0.10	0.91	-656
m3	4	1319	4.4	0.09	1.00	-655
m2	3	1348	33.1	0.00	1.00	-671
m0	2	1348	33.5	0.00	1.00	-672

Advantages of information criteria:

- You can compare all models that share the same null model.
- The computed values of AIC/BIC are independent from the order and number of models considered.
- The use of relative evidence and Akaike weights gives the reader greater insight into the relative merits of the competing models. In addition, Akaike weights quantify conclusions based on AIC analyses by specifying the amount of statistical confidence for the model with the lowest AIC value (Burnham & Anderson, 2002; Wagenmakers & Farrel, 2004).

Advantages of information criteria:

- You can compare all models that share the same null model.
- The computed values of AIC/BIC are independent from the order and number of models considered.
- The use of relative evidence and Akaike weights gives the reader greater insight into the relative merits of the competing models. In addition, Akaike weights quantify conclusions based on AIC analyses by specifying the amount of statistical confidence for the model with the lowest AIC value (Burnham & Anderson, 2002; Wagenmakers & Farrel, 2004).

Example:

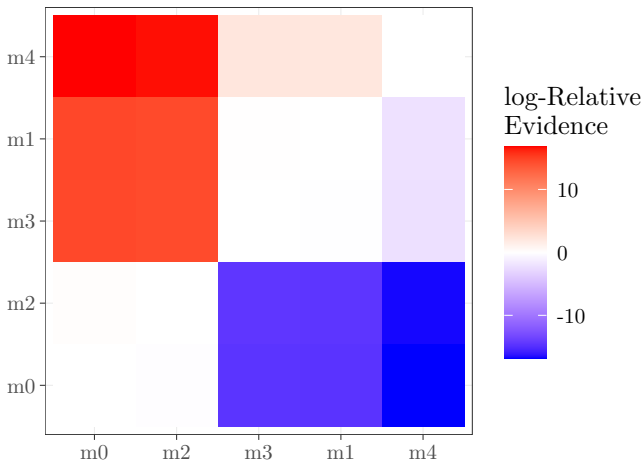
- Model m4 is about 10 times more likely than m3, and so on.

Pairwise comparisons

```
library(ADati)
plot_relative_evidence( AICtable$AICcWt,
                        labels = AICtable$Modnames )
```

Pairwise comparisons

```
library(ADati)
plot_relative_evidence( AICtable$AICcWt,
                        labels = AICtable$Modnames )
```



Used R packages

- `ADati`. Pastore M (2020).
- `AICcmodavg`. Mazerolle MJ (2023).
- `broom`. Robinson D, Hayes A, Couch S, Hvitfeldt E (2026).
- `formula.tools`. Brown C (2018).
- `ggplot2`. Wickham H (2016).
- `heplots`. Friendly M, Fox J, Monette G (2025).
- `knitr`. Xie Y (2025).
- `LEGIT`. Jolicoeur-Martineau A (2024).
- `modelbased`. Makowski D, Ben-Shachar M, Wiernik B, Patil I, Thériault R, Lüdecke D (2025).
- `patchwork`. Pedersen T (2025).
- `performance`. Lüdecke D, Ben-Shachar M, Patil I, Waggoner P, Makowski D (2021).
- `reghelper`. Hughes J, Beiner D (2023).



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